

DCI Final Project

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1 Objective

The objective of this report is to summarize the selected PCB stack-up, the component placement strategy, the routing approach and the main design-for-manufacturing decisions.

The final design is based on the STM32F746ZGTx microcontroller in an LQFP144 package. The PCB includes USB, Ethernet, two CAN interfaces, SPI flash memory, two analog input channels, GPIO connectors, a debug connector and the required power supply circuitry.

The goal was to keep the PCB manufacturable with standard LabCircuits capabilities while still providing good routing space, reliable signal return paths and suitable impedance-controlled routing for the critical interfaces.

2 Hardware Overview

The main functional blocks of the PCB are:

- STM32F746ZGTx microcontroller in LQFP144 package.
- USB OTG FS interface with Micro-AB connector and ESD protection.
- Ethernet interface using RMII signals, Ethernet PHY and RJ45 connector.
- Two CAN interfaces with CAN transceivers and differential CAN lines.
- SPI flash memory connected through SPI4.
- Two analog inputs connected to ADC3 channels.
- GPIO connector and SWD debug connector.
- Power supply from external DC input or USB VBUS.

The power supply uses two possible input sources. The external DC input is used as the preferred source, while USB VBUS can be used as a fallback source. A power multiplexer selects the active source for the 5 V rail. The 3.3 V rail is generated from the 5 V rail and supplies the MCU, Ethernet PHY and logic circuits.

3 Schematic Documentation

The schematic was organized hierarchically in Altium. The top-level schematic connects the functional blocks, including MCU, power supply, USB, Ethernet, CAN, ADC, SPI flash and connectors. This makes the design easier to read and keeps the individual schematic pages simple. The complete schematic is included as part of the project documentation. The most important schematic overview is shown in Figure 1.

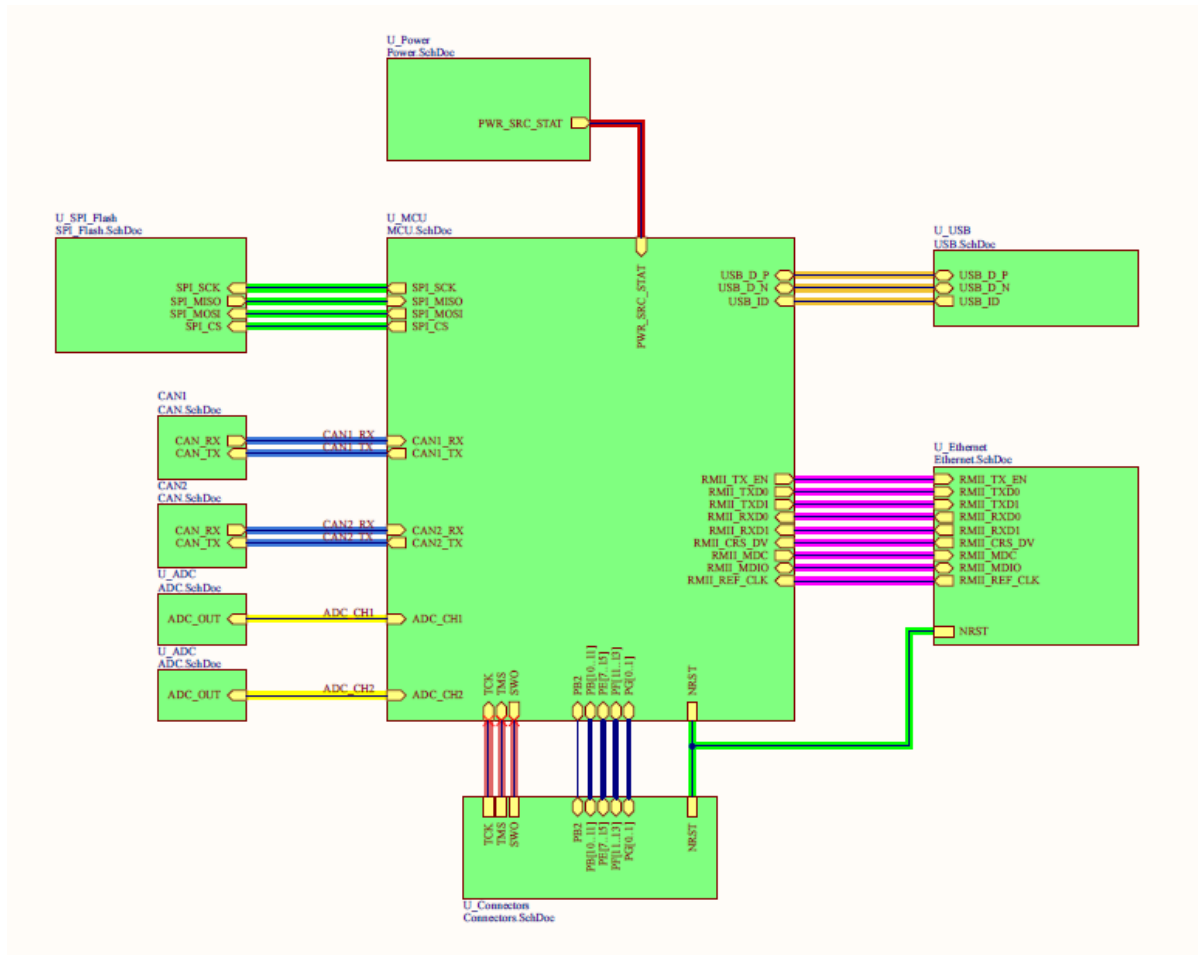


Figure 1: Top-level schematic showing the hierarchical project structure.

4 STM32CubeMX and IOC Configuration

The STM32CubeMX configuration file (`.ioc`) is included with the final project files. It documents the selected MCU, the pin assignment and the clock configuration used for the design.

The selected MCU is the STM32F746ZGTx in the LQFP144 package. The pinout was configured for the required project interfaces, including USB OTG FS, Ethernet RMII, CAN1, CAN2, SPI4, ADC3, SWD debug pins and GPIO signals.

The MCU pinout is shown in Figure 2. The clock tree is shown in Figure 3. The clock tree confirms the clock configuration used in CubeMX and provides the basis for the peripheral configuration.

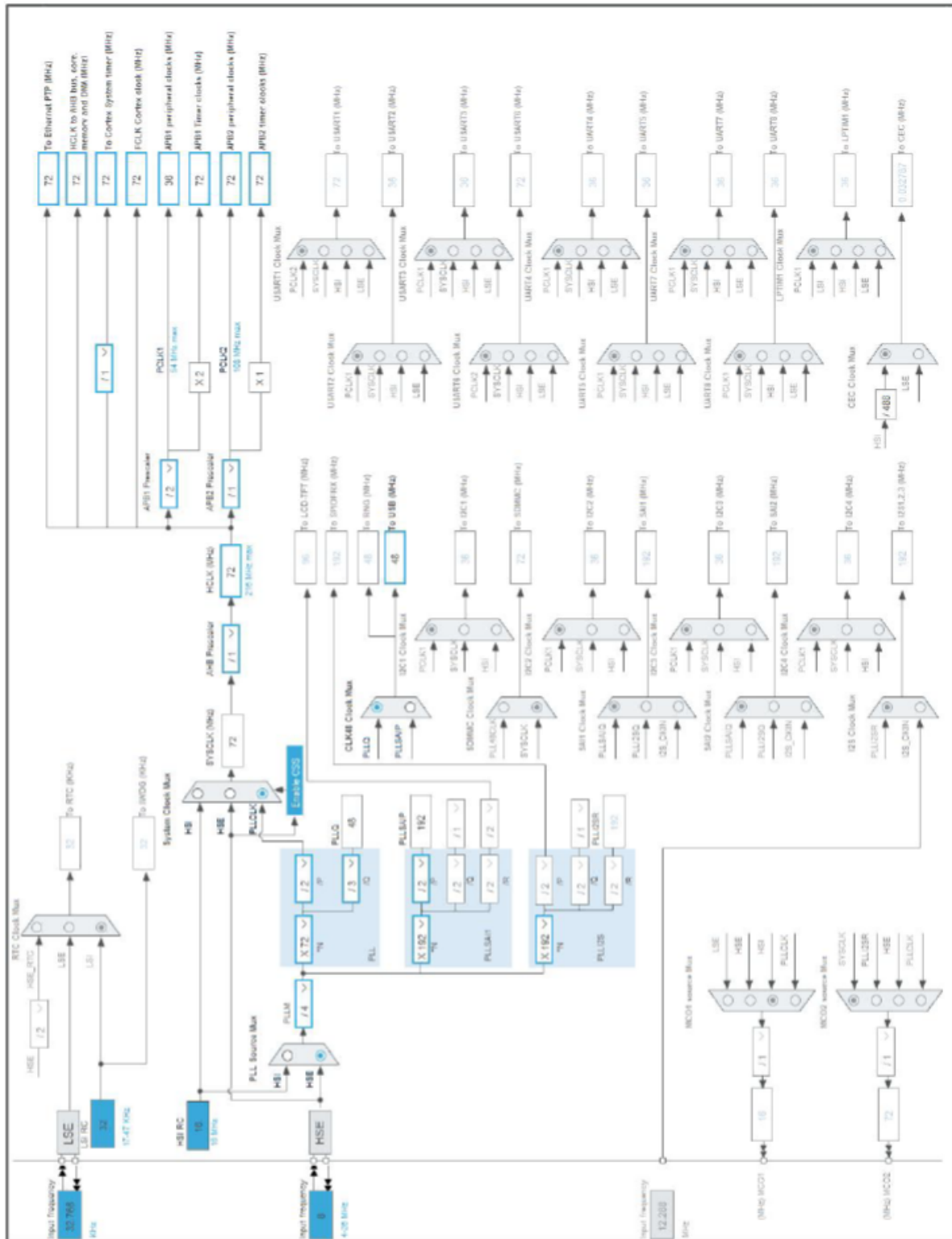


Figure 3: STM32CubeMX clock tree configuration.

5 PCB Stack-up

A four-layer PCB was selected as a compromise between cost, routing density and electrical performance. A two-layer board would be cheaper, but it would make routing and signal return paths more difficult. A six-layer board would provide more routing freedom, but the additional cost is not necessary for this project.

The selected layer structure is:

Signal Top (L1) – GND (L2) – Power (L3) – Signal Bottom (L4)

The top layer is used for the main components and for critical signal routing. The second layer is a continuous ground plane, which provides a good return path for signals routed on the top layer. The third layer is used for power distribution and is split into different voltage regions. The bottom layer is mainly used for secondary routing and less critical signals.

Because the power plane is split, the bottom layer does not always have an ideal continuous reference plane. Therefore, critical interfaces such as USB, Ethernet and CAN are preferably routed on the top layer, directly above the continuous ground plane.

The final stack-up has a total thickness of approximately 1.59 mm. The main stack-up is:

- Top Layer: signal layer, approximately 0.04 mm copper.
- Prepreg: approximately 0.18 mm.
- Internal Layer 1: continuous GND plane, approximately 0.04 mm copper.
- Core: approximately 1.00 mm.
- Internal Layer 2: power plane, approximately 0.04 mm copper.
- Prepreg: approximately 0.18 mm.
- Bottom Layer: signal layer, approximately 0.04 mm copper.

Standard FR-4 material was selected, since the frequencies in this project do not require special high-frequency material.

Layer Stack Legend						
	Material	Layer	Thickness	Dielectric Material	Type	Gerber
		Top Overlay			Legend	GTO
	Surface Material	Top Solder	0.03mm	SM-001	Solder Mask	GTS
	PbSn	Top Surface Finish	0.02mm		Surface Finish	
	CF-004	Top Layer	0.04mm		Signal	GTL
	Prepreg		0.18mm	PP-021	Dielectric	
	CF-004	Int1 (GND)	0.04mm		Internal Plane	GP1
	Core		1.00mm	Core-039	Dielectric	
	CF-004	Int2 (PWR)	0.04mm		Internal Plane	GP2
	Prepreg		0.18mm	PP-021	Dielectric	
	CF-004	Bottom Layer	0.04mm		Signal	GBL
	PbSn	Bottom Surface Finish	0.02mm		Surface Finish	
	Surface Material	Bottom Solder	0.03mm	SM-001	Solder Mask	GBS
		Bottom Overlay			Legend	GBO
Total thickness: 1.59mm						

Figure 4: Selected four-layer PCB stack-up in Altium.

6 Impedance Profiles and Design Rules

The required impedance profiles were created in the Altium Layer Stack Manager. These profiles are used to define routing rules for controlled-impedance traces and differential pairs.

A 50 Ohm single-ended profile was created for signals that require controlled impedance. Differential pair profiles were created for USB, Ethernet and CAN routing.

The final calculated impedance profiles are:

- Single-ended profile: target 50 Ohm, calculated 50.01 Ohm.
- USB differential pair: target 90 Ohm, calculated 89.98 Ohm.
- Ethernet differential pair: target 100 Ohm, calculated 100.02 Ohm.
- CAN differential pair: target 120 Ohm, calculated 119.97 Ohm.

The general design rules were defined according to LabCircuits Class 5. Minimum track width, clearance, via dimensions and solder-mask parameters were selected to stay within standard manufacturing limits. Wider traces were used for power nets where possible, while standard signal widths were used for general low-speed routing.

Dedicated net classes were created for different signal groups, for example power nets, differential pairs and general signals. This makes the routing rules easier to apply and reduces the risk of using unsuitable track widths or clearances.

7 LabCircuits Manufacturing Class

The PCB was designed according to **LabCircuits Class 5**. This class was selected because it offers sufficiently small track widths and clearances for the routing around the 144-pin LQFP microcontroller while still remaining a standard and cost-effective manufacturing option.

The selected stack-up, copper thicknesses, minimum track widths and clearances are compatible with this class. Therefore, no special high-cost manufacturing options are required.

8 Component Placement

The placement was developed using a functional-block approach. Components that belong to the same circuit function were placed close together. This reduces trace length and simplifies routing.

The microcontroller was placed in the right-central area of the board, which provides short connections to the surrounding interfaces. The Ethernet interface is placed close to the RJ45 connector, and the USB circuit is placed close to the USB connector. The two CAN circuits and analog input circuits are grouped close to their external connectors. This keeps the routing compact and reduces unnecessary crossing of signals.

The final board size is approximately 105 mm by 80 mm. The board outline is rectangular with rounded corners. Four mounting holes are placed close to the board corners.

Connectors such as USB, Ethernet, power input, screw terminals, GPIO and debug connector are placed near the board edges to allow easy access from outside the PCB.

The differential interfaces were placed so that USB, Ethernet and CAN signals can be routed as directly as possible. Critical components and signals were kept on the top layer whenever possible, because the top layer has the continuous ground plane directly underneath it.

Analog circuitry was kept away from noisy digital and power-switching regions. Instead of splitting the ground plane, the separation was mainly achieved by careful placement and routing. This keeps the ground plane continuous and avoids unnecessary interruptions of signal return paths.

Decoupling capacitors were placed as close as possible to their corresponding IC supply pins. Where possible, capacitors were placed directly below or close to the related ICs to reduce parasitic inductance and improve high-frequency decoupling.

Some low-speed support circuitry was placed on the bottom layer to reduce routing congestion on the top layer.

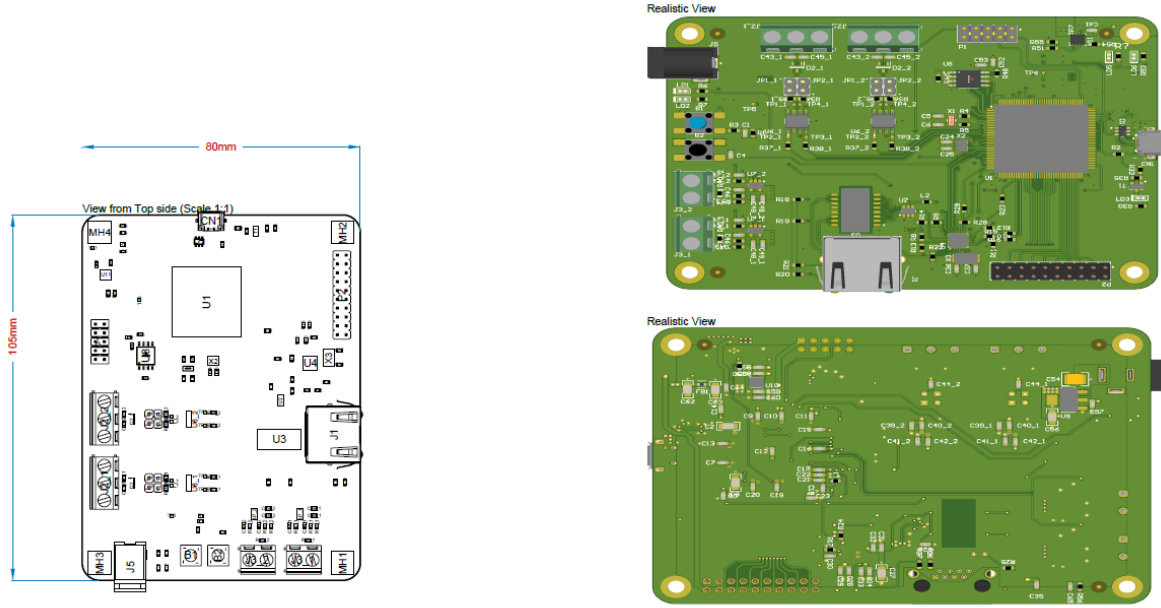


Figure 5: Final PCB component placement.

9 Routing and DFM Considerations

The routing was done with standard manufacturing methods in mind. The goal was to avoid unnecessary complexity and to keep the board suitable for reliable fabrication and assembly.

9.1 Board Outline and Connectors

The PCB uses a simple rectangular board outline with rounded corners. This avoids unnecessary milling complexity. Mounting holes are placed near the corners with sufficient clearance to surrounding components.

The main connectors are placed near the board edges. This makes external cabling easier and improves mechanical accessibility.

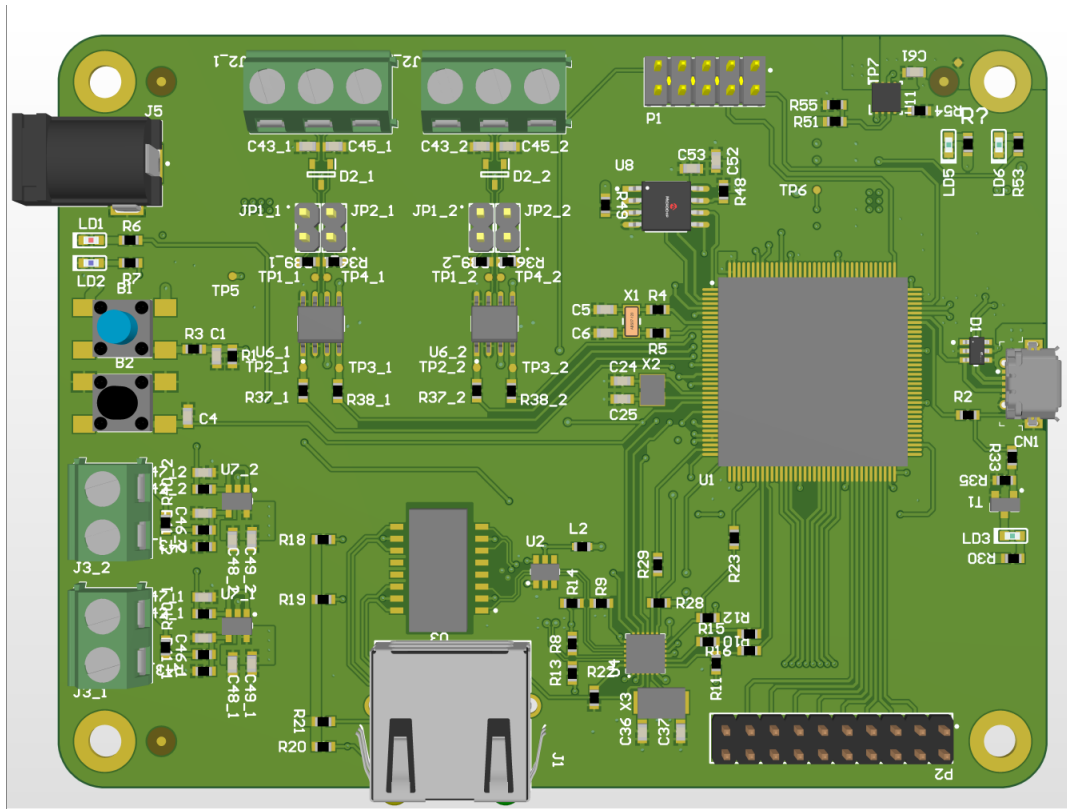


Figure 6: Final routed PCB layout, top side.

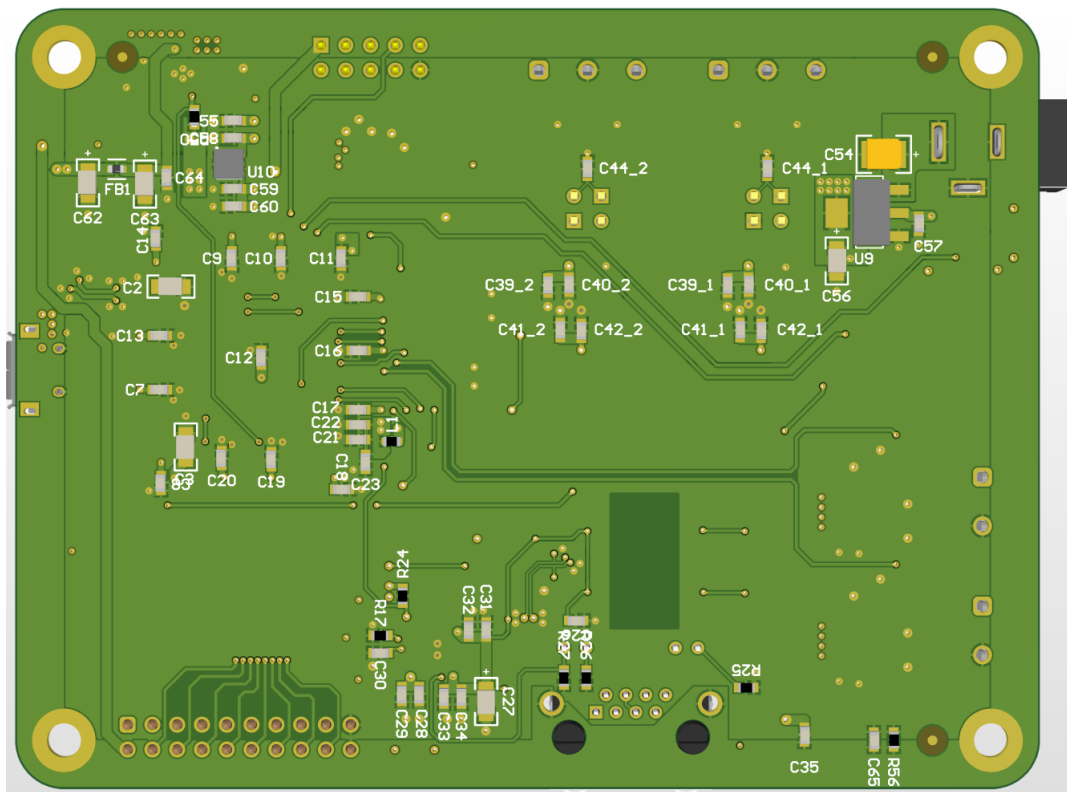


Figure 7: Final routed PCB layout, bottom side.

9.2 Via Technology

Only standard through-hole vias were used. No microvias, blind vias or buried vias are required. This keeps the manufacturing process simple and avoids additional cost.

9.3 Symmetric Stack-up

The four-layer stack-up was designed to be symmetric. A symmetric stack-up reduces the risk of board warpage during lamination and soldering. This improves mechanical stability and assembly reliability.

9.4 Copper Pours

Ground copper pours were added on the outer layers. They help to create a more uniform copper distribution and provide additional ground connection areas. A more uniform copper distribution also helps during manufacturing, because it reduces the risk of uneven etching or plating.

9.5 Thermal Reliefs

Thermal reliefs were used for pads connected to copper planes. This makes soldering easier because the pad is not directly connected to a large copper area. Without thermal reliefs, the copper plane could remove too much heat from the pad, which may lead to poor solder joints.

9.6 Trace Routing

Traces were routed mainly with 45-degree angles. Sharp 90-degree corners were avoided because they are not good practice for PCB routing and can create unnecessary manufacturing and signal-integrity issues.

Differential pairs were routed with controlled width and spacing according to the calculated impedance profiles. Unnecessary vias and layer changes were avoided on critical signal paths where possible.

9.7 Teardrops

Teardrops were added at via-to-trace and through-hole pad-to-trace transitions. This increases the copper area at these transitions and makes the connection more robust against drilling and etching tolerances.

9.8 Via Tenting

Vias were tented on the top side. The bottom side was left open to avoid trapping gases during manufacturing. This reduces the risk of solder mask defects such as blistering.

9.9 Fiducials

Board-level fiducials were placed on the outer layers. They are used by pick-and-place machines as optical reference points during automated assembly.

10 Critical Analysis

The main advantage of the design is the continuous ground plane directly below the top layer. This provides a good return path for critical signals and improves signal integrity. The four-layer stack-up also gives enough routing space while keeping the cost lower than a six-layer board.

Another advantage is that only standard manufacturing technologies are used. The design uses LabCircuits Class 5, standard FR-4 material and standard through-hole vias. No HDI technologies are required.

The main weakness is that the bottom layer uses the split power plane as its reference. Therefore, it is not ideal for critical signals. This was mitigated by placing and routing important interfaces on the top layer whenever possible.

The board size of approximately 105 mm by 80 mm provides enough routing space for the MCU, connectors and interfaces. A smaller PCB would reduce manufacturing cost, but it would also make the placement and routing around the 144-pin MCU, Ethernet interface and connectors more difficult.

Overall, the selected stack-up, placement and routing strategy provide a practical compromise between cost, manufacturability and electrical performance.

11 Conclusion

A four-layer PCB with the structure Signal – GND – Power – Signal was selected. This provides a good balance between cost and performance. The continuous ground plane below the top layer is the most important feature for signal integrity.

The final stack-up has a total thickness of approximately 1.59 mm and uses standard FR-4 based materials. The required impedance profiles for 50 Ohm single-ended routing and 90 Ohm, 100 Ohm and 120 Ohm differential routing were created in Altium and used for the corresponding routing rules.

The final design is based on the STM32F746ZGTx microcontroller and includes USB, Ethernet, two CAN interfaces, SPI flash, ADC inputs, GPIO and power management. The IOC file is included with the final project documentation and documents the MCU pinout and clock tree configuration. The schematic documentation is also included to show the electrical structure of the design.

The PCB was designed according to LabCircuits Class 5 and uses standard manufacturing options. The routing avoids unnecessary complexity and includes several DFM measures such as a simple board outline, standard vias, thermal reliefs, teardrops, copper pours and fiducials.

References

- Lab Circuits, ‘4 Layers, 1.6 mm, Build-up 1’, available at: https://www.lab-circuits.com/uploads/doc/multicapa/4layers_1.6mm_LabCode1.pdf
- Lab Circuits, ‘Fabricacion multicapa – parametros de fabricacion’, available at: <https://www.lab-circuits.com/es/fabricacion-multicapa>
- Diseno de Circuitos Impresos, UPV course slides.
- STMicroelectronics, STM32F746ZGTx datasheet and STM32CubeMX configuration report.
- Altium Designer PCB schematic, fabrication and assembly output files of the final project.